HR ATTRITION ANALYTICS

Why Employees Leave

An end-to-end HR attrition analytics project using Python, PostgreSQL, and Power BI.

1 in 6

employees left.

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THE STARTING POINT

Attrition affects 1 in 6 employees

1,470

employees across 35 fields

237

employees who left

16.1%

overall attrition

Business question

Which employee groups need the most focused retention action?

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FROM DATA TO ACTION

The analysis moved from raw data to action

01

Prepare

Cleaned and prepared the dataset in Python.

02

Question

Answered business questions with PostgreSQL.

03

Explain

Built measures, charts, and slicers in Power BI.

04

Act

Translated patterns into HR recommendations.

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DATA QUALITY FIRST

Data quality was established before analysis

Verified

No missing values and no duplicate rows were found.

Simplified

Constant fields and employee ID were removed.

Engineered

Created an attrition flag plus age, income, and tenure bands.

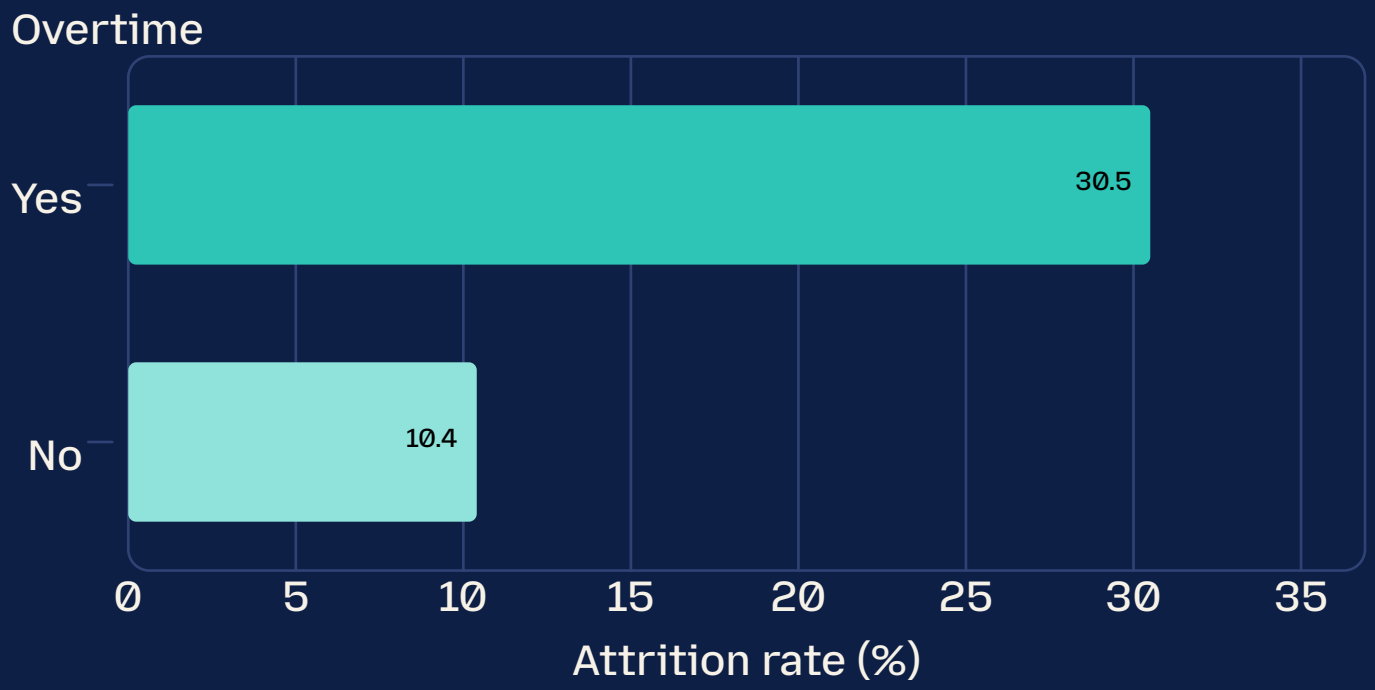
Compared fairly

Used rates instead of counts to compare groups fairly.

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STRONGEST OBSERVED SIGNAL

Overtime is the clearest attrition signal



≈3×

The overtime group leaves at roughly three times the rate.

Action: review workload, staffing, and overtime policy.

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DECISION SUPPORT

The Power BI dashboard turns analysis into a decision tool

1,470

KPI: employees

237

KPI: employees left

16.1%

KPI: attrition rate

6,833

KPI: average monthly income

Breakdowns: Department · job role · overtime · age · tenure · income

Filters: Department · gender · overtime

Decision value: The dashboard supports targeted HR conversations instead of one-size-fits-all action.

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WHERE TO FOCUS

Recommendations focus on the highest-risk groups

Overtime:	Rebalance workloads and review staffing needs
Young and new employees:	Add mentoring, career paths, and first-year check-ins
Low-income roles:	Review pay bands and progression opportunities
Sales Representatives:	Review targets, incentives, and workload
Frequent travellers:	Offer flexibility and reduce avoidable travel

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THE LIMIT OF THIS ANALYSIS

The next step is moving from association to prediction

What the analysis shows

Association

This analysis identifies associations; it does not prove that one factor causes attrition.

Some groups are small, so their rates are less stable.

What comes next

Prediction

Add job satisfaction and work-life balance views.

Compare factors together and test a logistic regression model.

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PORTFOLIO TAKEAWAY

What this project demonstrates

End-to-end ownership

From data cleaning to business recommendations.

Practical technical skills

Python, PostgreSQL, DAX, and Power BI.

Analytical judgment

Choose rates over misleading counts.

Business translation

Turn analysis into clear, prioritised action.

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